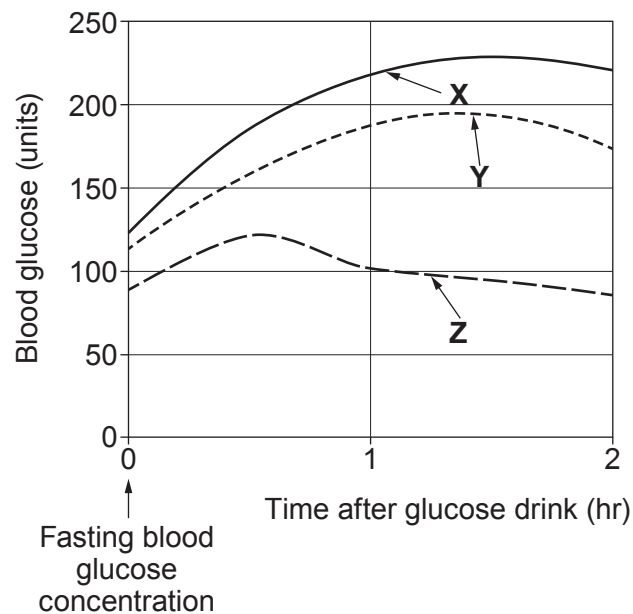


4. Medical scientists have developed methods to diagnose, monitor and treat diseases.

- (a) For one type of diabetes test, you fast (don't eat) overnight, and the fasting blood glucose level is measured. Then you drink a glucose drink, and blood glucose levels are tested for the next two hours. A blood glucose level less than 140 units is normal. A reading of more than 200 units after two hours indicates diabetes. A reading between 140 and 199 units indicates prediabetes.



(i) State which line, **X**, **Y** or **Z** would show a person is diabetic. [1]

(ii) State **one** method used to control diabetes. [1]

.....



- (b) Ionising radiation and ultrasound are used by medical scientists. Examples of imaging techniques are shown below. Complete the table. [4]

	Imaging technique	Example of its medical use
A		X-ray detects broken bones
B		
C		

- (c) Damage to DNA can lead to cancer.

- (i) Complete the sentence below to show the base pairs found in DNA. [1]

T pairs with and G pairs with

- (ii) Give **one** reason why imaging technique **A** could cause DNA damage. [1]

.....

